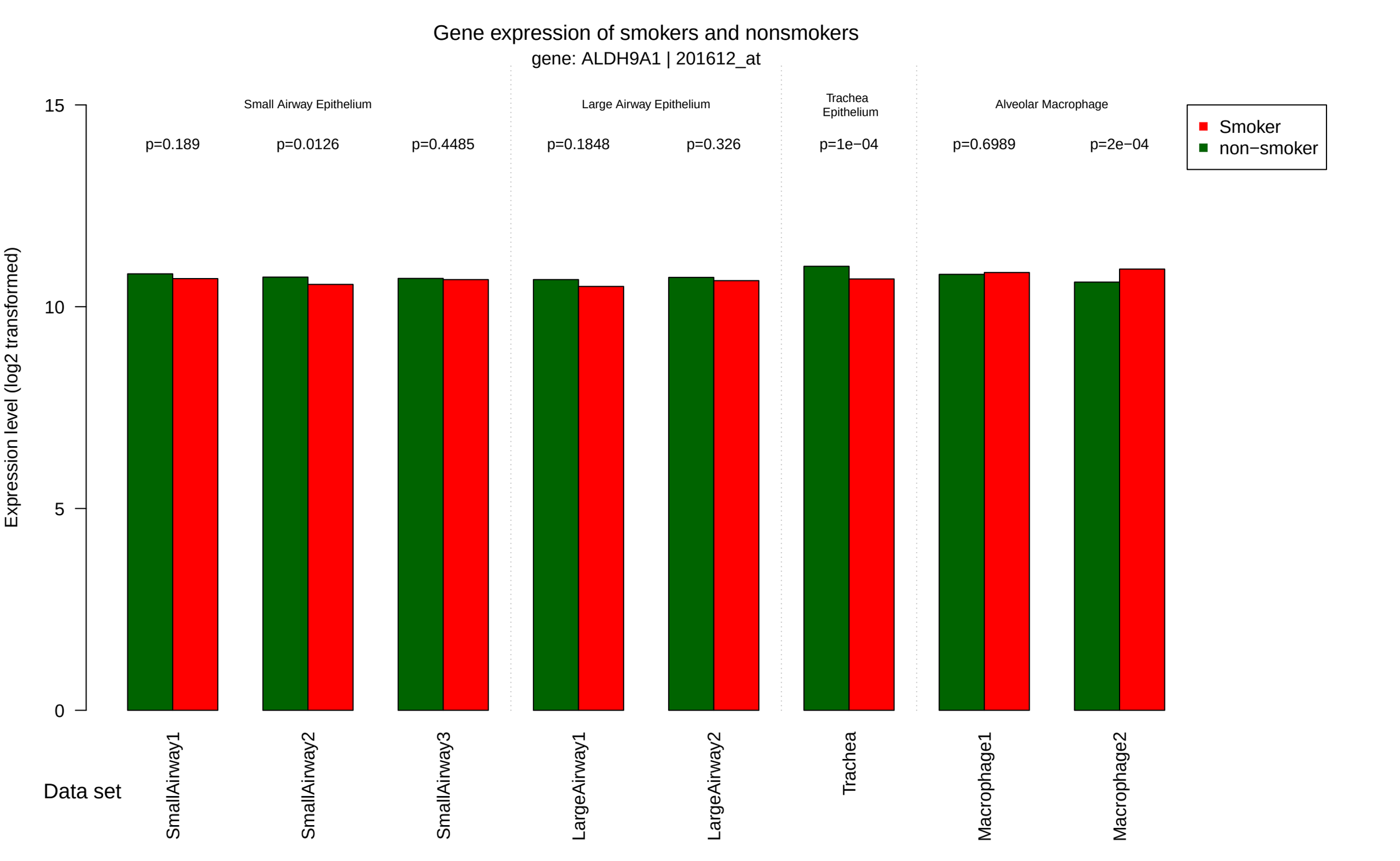
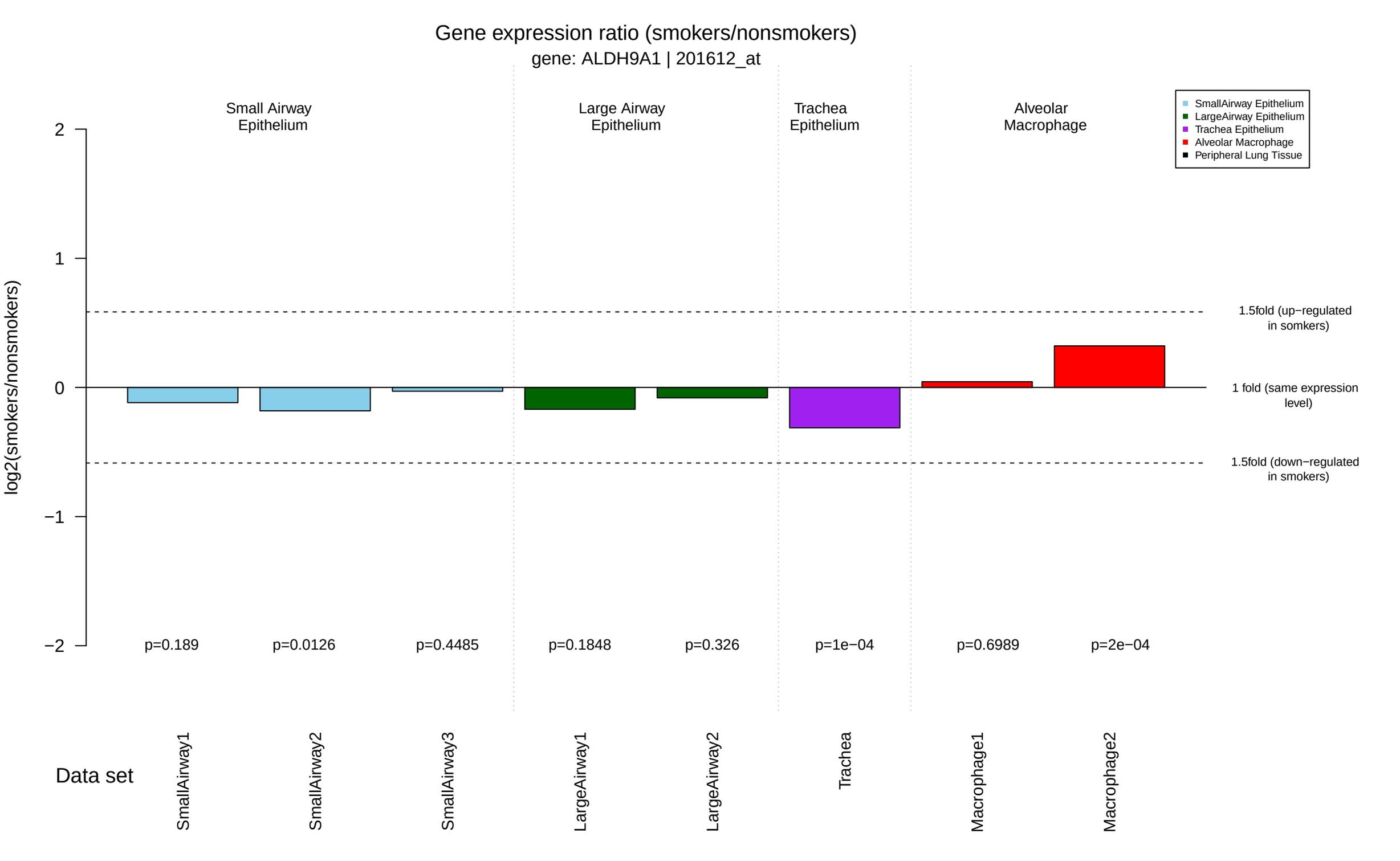
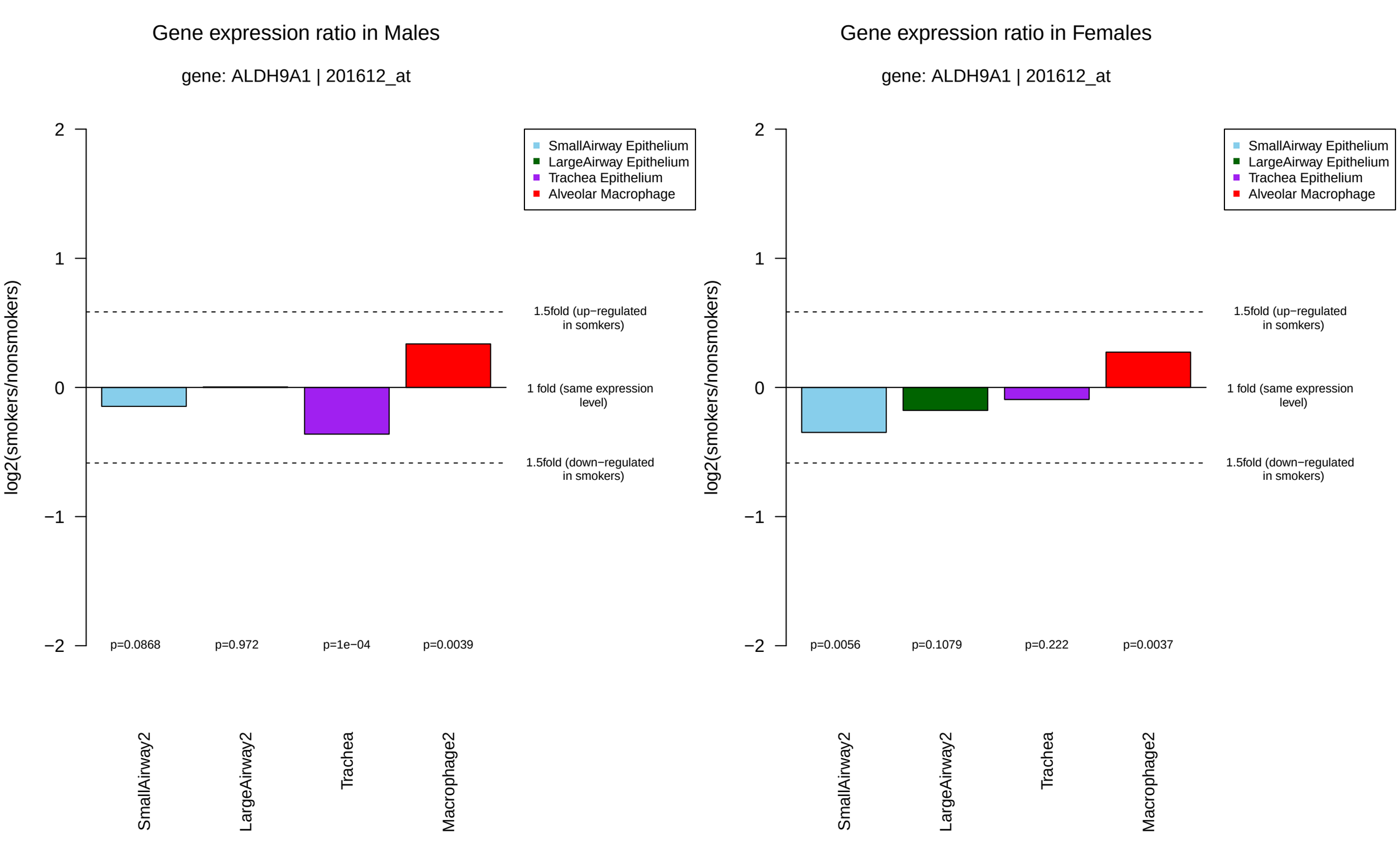
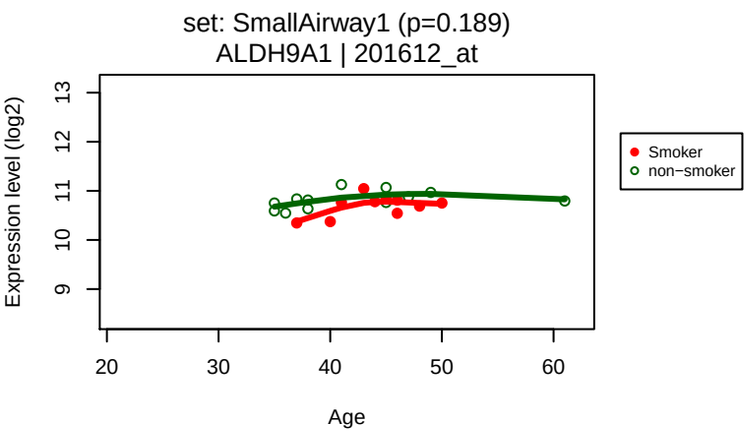
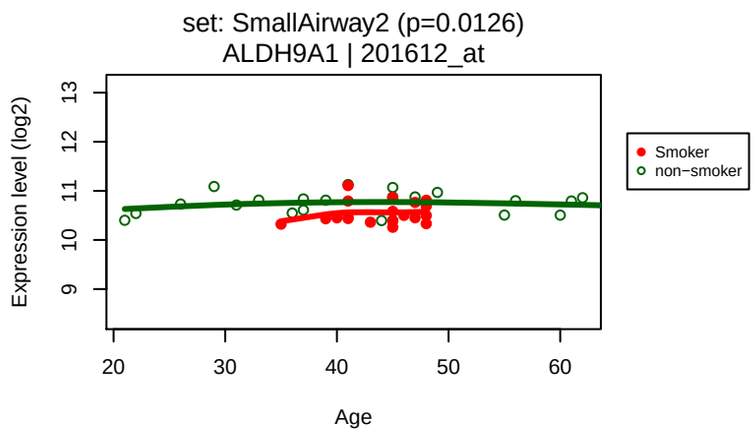
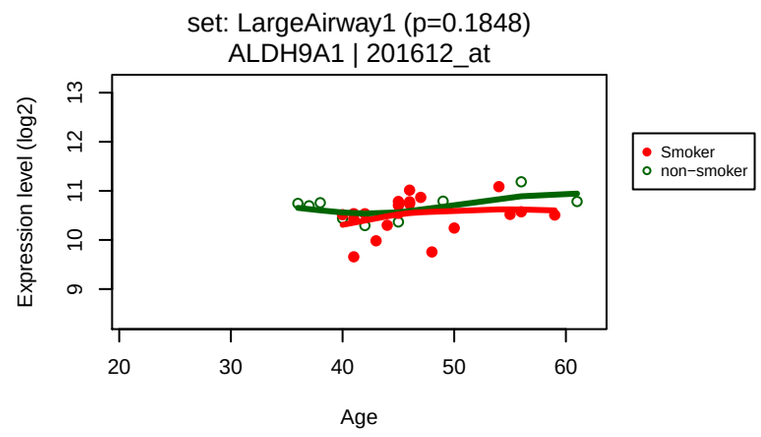

Gene expression of smokers and nonsmokers
gene: ALDH9A1 | 201612_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.189
p=0.0126
p=0.4485
p=0.1848
p=0.326
p=1e−04
p=0.6989
p=2e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

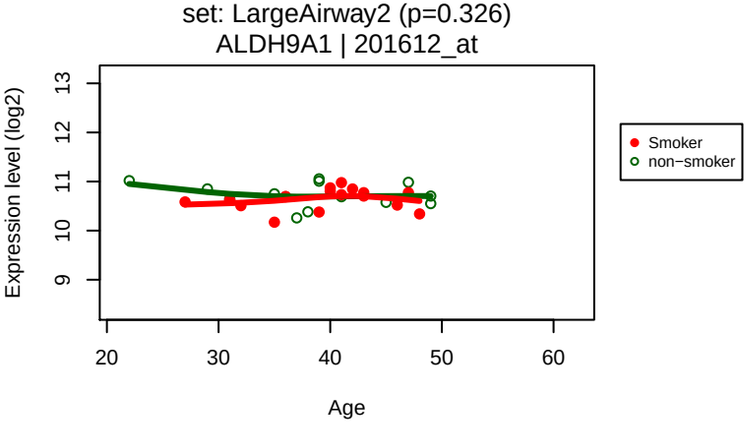

Gene expression ratio (smokers/nonsmokers)
gene: ALDH9A1 | 201612_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.189
p=0.0126
p=0.4485
p=0.1848
p=0.326
p=1e−04
p=0.6989
p=2e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

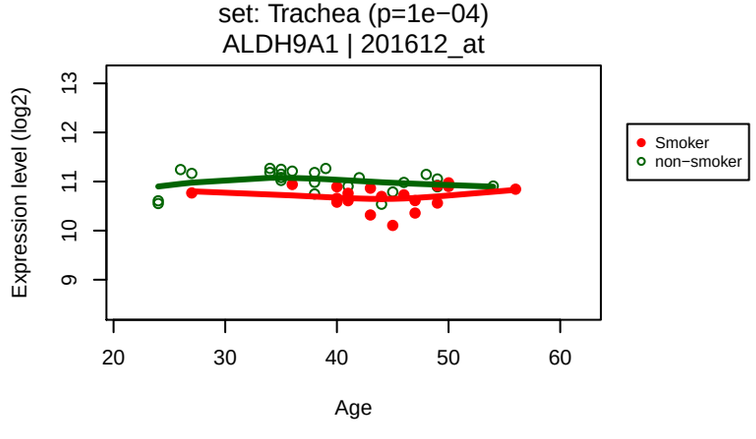

Gene expression ratio in Males
gene: ALDH9A1 | 201612_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0868
p=0.972
p=1e−04
p=0.0039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH9A1 | 201612_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0056
p=0.1079
p=0.222
p=0.0037
SmallAirway2
LargeAirway2
Trachea
Macrophage2

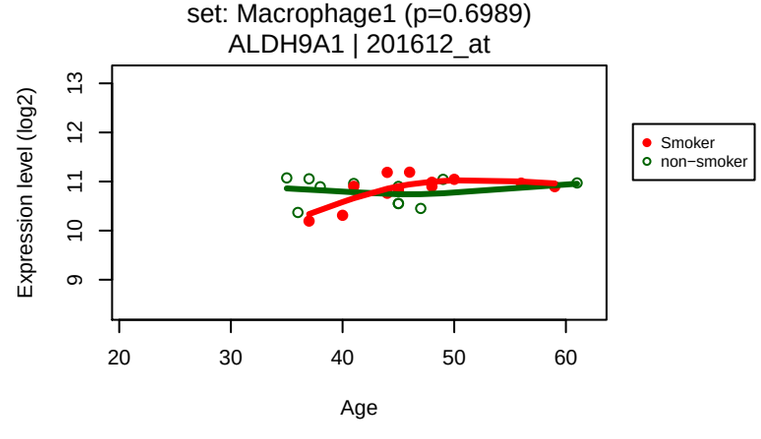

set: SmallAirway1 (p=0.189)
ALDH9A1 | 201612_at
Expression level (log2)
Age
Smoker
non−smoker

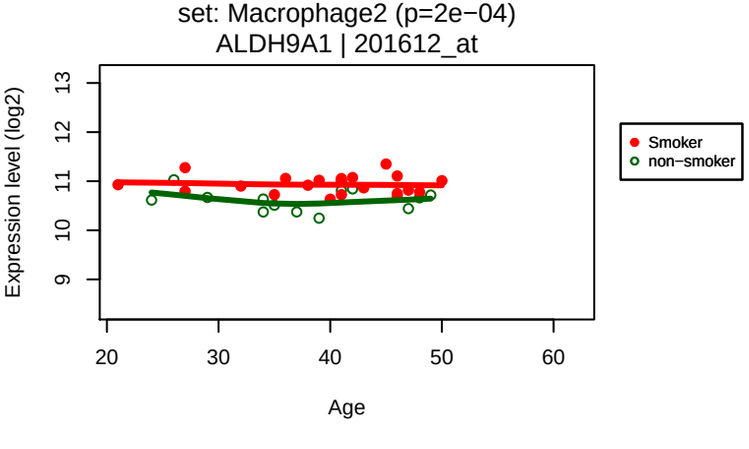

set: SmallAirway2 (p=0.0126)
ALDH9A1 | 201612_at
Expression level (log2)
Age
Smoker
non−smoker

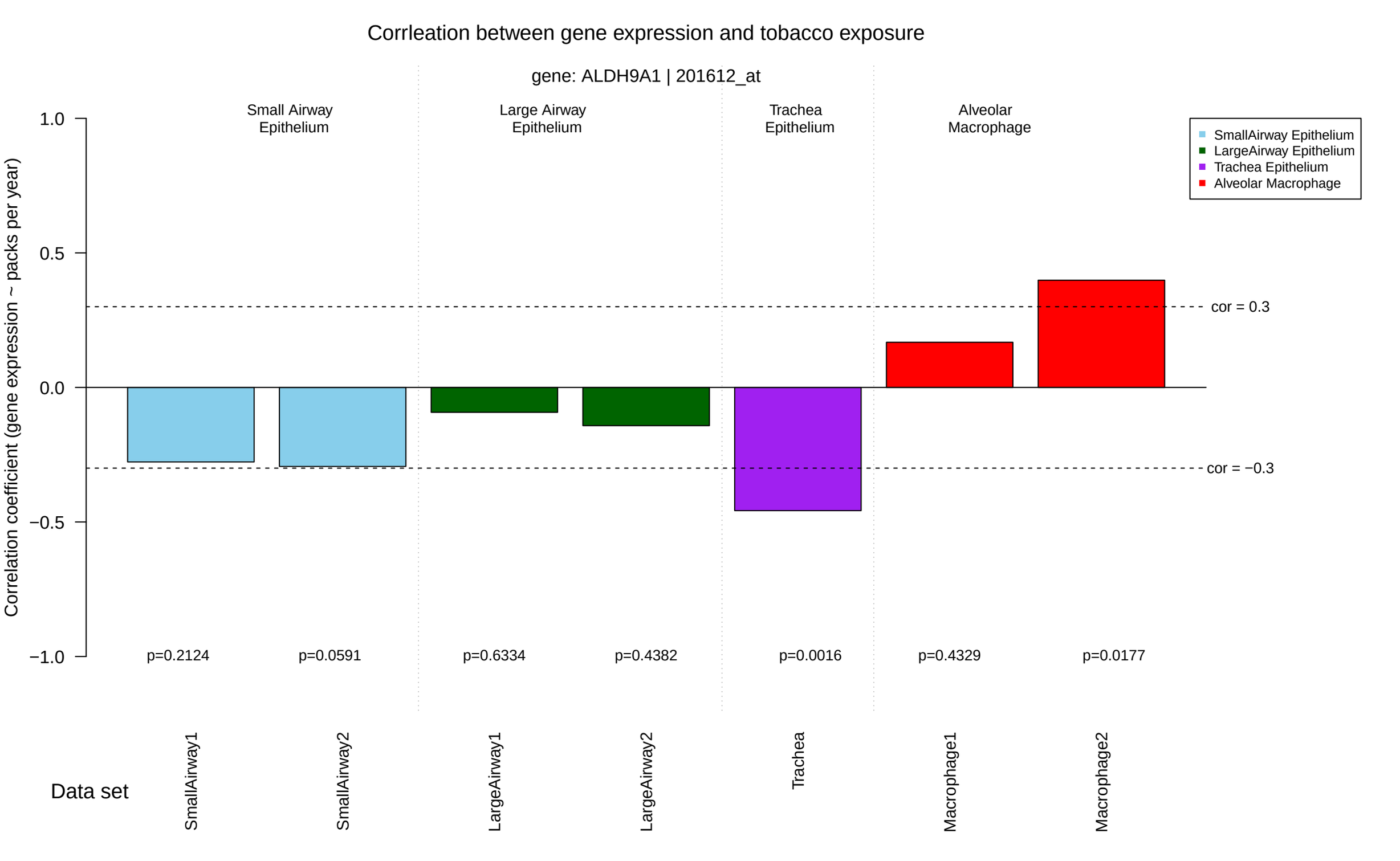

set: LargeAirway1 (p=0.1848)
ALDH9A1 | 201612_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.326)
ALDH9A1 | 201612_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
ALDH9A1 | 201612_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6989)
ALDH9A1 | 201612_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=2e−04)
ALDH9A1 | 201612_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH9A1 | 201612_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2124
p=0.0591
p=0.6334
p=0.4382
p=0.0016
p=0.4329
p=0.0177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2